BLL

BLL[23]

1. Collect Layer Activations

- Query LLM on training questions
- Extract layer activations $h^{(l,i)}$
- Compute activation change:

$$y^{(l,i)} = h^{(l,i)} - h^{(l-1,i)}$$

- Split into **Cor** (correct) and **Incor** (incorrect)

2. Bayesian Linear Modeling (Per Neuron, Per Mode)

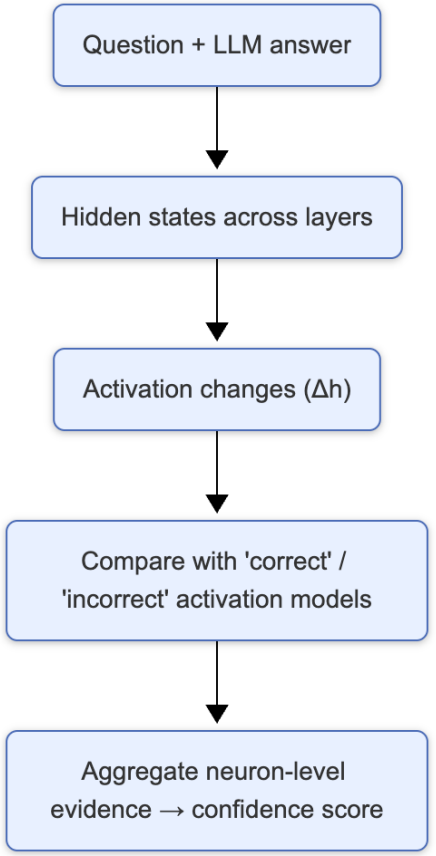
For each neuron (l, i), fit **two Bayesian linear models**:

$$y^{(l,i)} = w_{u,0}^{(l,i)} + w_u^{(l,i)}\, h^{(l-1,:)} + \varepsilon_u^{(l,i)}, \quad \varepsilon_u \sim \mathcal{N}(0, \sigma_u^2)$$

- $u = 1$: Correct mode
- $u = 0$: Incorrect mode
- Obtain posterior predictive:

$$p\left(y^{(l,i)} \mid h^{(l-1,:)}, u\right)$$

3. Posterior → Global Confide



BLL extracts two feature types:

(A) Posterior log-likelihood

$$-\log p\left(y^{(l,i)} \mid u\right)$$

(B) Likelihood Ratio (LLR)

$$\mathrm{LLR}^{(l,i)} = \log \frac{p\left(y^{(l,i)} \mid u = 1\right)}{p\left(y^{(l,i)} \mid u = 0\right)}$$

- Indicates whether the activation **resembles the correct or incorrect mode**
- All neuron-level features combined via **Elastic-Net probe** → final confidence score

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- **Task Type:** Open-ended generation

- **Introduction:** BLL fits Bayesian linear models to layer-wise activation differences conditioned on correctness and aggregates posterior features through sparse regression to estimate response reliability. It quantifies decoding uncertainty and provides a structured way to extract distributional signals from intermediate activations for factual uncertainty assessment.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Single-pass	Decoding	White-box	Fine-tuning	High	Ranking/Calibration

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